

3. (a) (i) Friction  $f$  between the tyres and the road.

1A

$$f = \frac{mv^2}{r}$$

1M

$$8000 = \frac{1200 v^2}{45 \text{ m}}$$

$$v = 17.3 \text{ m s}^{-1}$$

1A 3

- (ii) Smaller

1A

For the same  $f$ ,  $v^2 \propto r$ ; when  $r$  decreases  $v$  decreases.

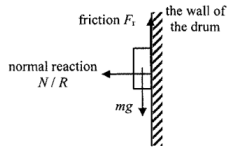
1A 2

- (b) (Max) friction / coefficient of friction reduced,  
not enough to provide the centripetal force / acceleration required for circular motion.  
Or Tracking or allowed speed lowered.

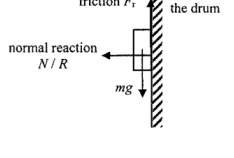
1A

1A 2



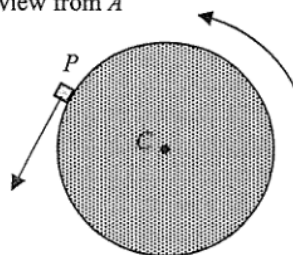
|                                                                            |                                                                                                                                                                                                                                                                              |    |   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
| 4. (a) (i)                                                                 |                                                                                                                                                                                             | 2A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (ii) Normal reaction (force acting on drum/wall of the drum by the object) |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 1 |
| (b) (i)                                                                    | $\omega = 78.5 \text{ rad s}^{-1}$<br>rotation speed = $\frac{78.5}{2\pi} \times 60$<br>$= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$                                                                                                                             | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            | <div style="border: 1px solid black; padding: 5px;"> Or<br/> <math>T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}</math> and<br/> <math>f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}</math><br/> <math>= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}</math> </div> | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (ii) centripetal force                                                     |                                                                                                                                                                                                                                                                              |    |   |
|                                                                            | $= m\omega^2 r$<br>$= (2 \times 10^{-5})(78.5)^2(0.25)$<br>$= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$                                                                                                                                                                 | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (iii) $v = r\omega$                                                        |                                                                                                                                                                                                                                                                              |    |   |
|                                                                            | $= 0.25 \times 78.5$<br>$= 19.625 \approx 19.6 \text{ m s}^{-1}$                                                                                                                                                                                                             | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (iv) I                                                                     |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 1 |

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|                                                                            |                                                                                                                                                                                                                                                                              |    |   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
|                                                                            |                                                                                                                                                                                           | 2A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (ii) Normal reaction (force acting on drum/wall of the drum by the object) |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 1 |
| (b) (i)                                                                    | $\omega = 78.5 \text{ rad s}^{-1}$<br>rotation speed = $\frac{78.5}{2\pi} \times 60$<br>$= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$                                                                                                                             | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            | <div style="border: 1px solid black; padding: 5px;"> Or<br/> <math>T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}</math> and<br/> <math>f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}</math><br/> <math>= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}</math> </div> | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (ii) centripetal force                                                     |                                                                                                                                                                                                                                                                              |    |   |
|                                                                            | $= m\omega^2 r$<br>$= (2 \times 10^{-5})(78.5)^2(0.25)$<br>$= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$                                                                                                                                                                 | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (iii) $v = r\omega$                                                        |                                                                                                                                                                                                                                                                              |    |   |
|                                                                            | $= 0.25 \times 78.5$<br>$= 19.625 \approx 19.6 \text{ m s}^{-1}$                                                                                                                                                                                                             | 1M |   |
|                                                                            |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 2 |
| (iv) I                                                                     |                                                                                                                                                                                                                                                                              | 1A |   |
|                                                                            |                                                                                                                                                                                                                                                                              |    | 1 |

4. (a) (i)  $F = m\omega^2 r = 0.020 \times 6^2 \times 0.50$   
 $= 0.36 \text{ N}$

(ii) Top view from A



(b) (i) same angular speed

(ii) different / smaller magnitude of centripetal acceleration  
 $(a_Q < a_P)$   
 as Q takes a different / smaller radius ( $r_Q < r_P$ )

|    |
|----|
| 1M |
| 1A |
| 2  |
| 1A |
| 1  |
| 1A |
| 1  |
| 1A |
| 1A |
| 2  |